

Multivariate calculus

Math4AI

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Machine Learning 1

Introduction



If you fall, I'll be there - Floor

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Welcome to the Math4AI site associated with the MSc in Artificial Intelligence from the University of Amsterdam. This site is provided to help you familiarize yourself with the mathematics needed to become proficient in machine learning.

Essentials

The essential mathematics any student of machine learning should know are **linear algebra**, **multivariate calculus**, and **statistics**. These three areas of mathematics truly form the foundation of all topics in machine learning. That is why our 'essentials' sections are dedicated to covering the most important topics within these areas for our purposes:

- **Linear algebra**, studying how to represent, transform, and analyze high dimensional data as vectors.
- **Multivariate calculus**, studying how to do differential calculus in high dimensional spaces.
- **Statistics**, studying how to analyze data, quantify uncertainty, and describe random events.

Each section contains a tutorial on top and a summary at the bottom discussing the highlights of the section, and one important exercise to help you understand the main point of the section. Please notice the search bar at the top if you want to quickly look for a topic (e.g. 'marginalization').

If you see any errors on this site, please [write us](#) and we will make sure to address them as soon as possible.

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Check the website math4.ai (click!)

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What will we cover?

You already know derivatives of functions $f: \mathbb{R} \rightarrow \mathbb{R}$, such as $f(x) = x^2$.

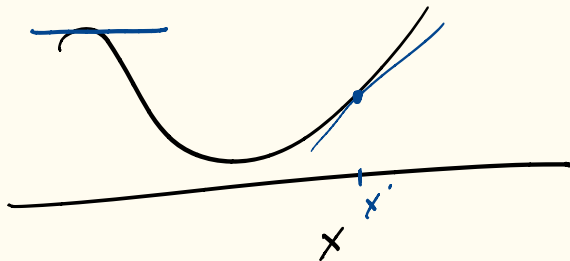
Today, we extend this to **multiple inputs and outputs**: $\mathbf{f}: \mathbb{R}^m \rightarrow \mathbb{R}^n$.

We will use **index notation** to differentiate common ML functions. Many have simple derivatives, which makes large models efficient to train.

What is a derivative?

A derivative measures **how sensitive f is to a change in x** . If the input changes by a tiny amount Δx , how much does the output change relative to Δx ?

It also gives the best local linear approximation of f :



Why do we care?

A model has parameters θ . A loss $\mathcal{L}$ measures its error on a dataset $\mathcal{D}$:

$$\mathcal{L} : \Theta \rightarrow \mathbb{R},$$

We want parameters that minimize the loss. The gradient tells us how the loss changes as the parameters change.

Small example

Linear regression predicts a scalar target y from $\mathbf{x} \in \mathbb{R}^m$, using parameters $\mathbf{w}$:

$$\hat{y} = \underline{\mathbf{w} \cdot \mathbf{x}} = \sum_{i=1}^m w_i x_i.$$

For a dataset $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$, the **mean squared error** is the average squared difference between predictions and targets:

$$\mathcal{L}_{\text{MSE}}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (\mathbf{w} \cdot \mathbf{x}_n - y_n)^2.$$

Our goal is to find $\mathbf{w}$ with the smallest loss by solving $\frac{d}{d\mathbf{w}} \mathcal{L}_{\text{MSE}}(\mathbf{w}) = \mathbf{0}$.

We will return to linear regression next week.

Notation

For $\mathbf{f}: \mathbb{R}^m \rightarrow \mathbb{R}^n$, differentiate each output f_i with respect to each input x_j :

$$\left[\frac{\partial f_i}{\partial x_j} \right]$$

The **Jacobian**¹ collects all $n \times m$ partial derivatives: $\left[\frac{df}{dx} \right]_{ij} = \frac{\partial f_i}{\partial x_j}$

$$\frac{d\mathbf{f}}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_m} \end{bmatrix}.$$

¹Jacobi: "Mathematics is the science of what is clear by itself", what would he think of this lecture?

Visualizing functions

Let's start with basic functions from $f: \mathbb{R} \rightarrow \mathbb{R}$, e.g. $y = x^2$.

$$\boxed{x} \longrightarrow x^2 \longrightarrow \boxed{y}$$

$$\frac{dy}{dx} = \frac{d}{dx} (x^2) = 2x$$

Visualizing functions

Tiny bit more spicy: $f: \mathbb{R} \rightarrow \mathbb{R}$, where $y = \exp(\sin(x))$.

$$\boxed{x} \longrightarrow \sin x \longrightarrow \boxed{u} \longrightarrow \exp u \longrightarrow \boxed{y}$$

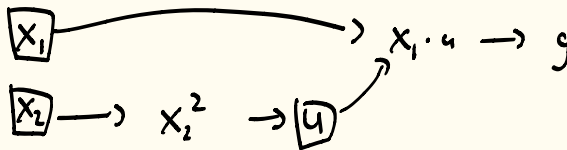
$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} = \cos x \cdot \exp u \\ &= \cos x \exp(\sin x) \end{aligned}$$

Visualizing functions

Let's go one step further, and consider a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$f: (x_1, x_2) \mapsto x_1 x_2^2.$$

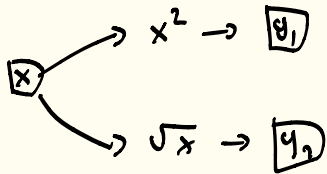
$$\frac{dy}{d\bar{x}} = \begin{bmatrix} \frac{\partial y}{\partial x_1} & \frac{\partial y}{\partial x_2} \end{bmatrix}$$


$$x_1 \cdot u \rightarrow y = \begin{bmatrix} x_1^2 & 2x_1 x_2 \end{bmatrix}$$

$$\frac{dy}{dx_1} = u = x_2^2 \quad \Bigg| \quad \frac{dy}{dx_2} = \frac{dy}{du} \cdot \frac{du}{dx_2} = 2x_1 x_2$$

Visualizing functions

We can also have a function $\mathbf{f} : \mathbb{R} \rightarrow \mathbb{R}^2$ which maps $\mathbf{f} : x \mapsto \begin{bmatrix} x^2 \\ \sqrt{x} \end{bmatrix}$.



$$\frac{d\mathbf{f}}{dx} = \begin{bmatrix} \frac{df_1}{dx} \\ \frac{df_2}{dx} \end{bmatrix}$$

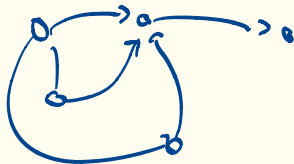
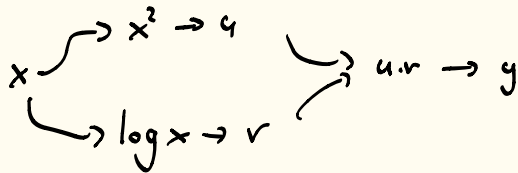
$$\frac{dy_1}{dx} = 2x$$

$$\frac{dy_2}{dx} = \frac{1}{2\sqrt{x}}$$

$$= \begin{bmatrix} 2x \\ \frac{1}{2\sqrt{x}} \end{bmatrix}$$

Visualizing functions

What about $f: x \mapsto x^2 \cdot \log x$?



$$\frac{dy}{dx} = \frac{\partial y}{\partial u} \cdot \frac{du}{dx} + \frac{\partial y}{\partial v} \cdot \frac{dv}{dx}$$

$$= v \cdot 2x + u \cdot \frac{1}{x}$$

$$= 2x \log x + x = x(2 \log x + 1)$$

Differentiating a dot product

Many ML operations use linear algebra: their derivatives are simple and efficient to compute.

Consider $f(\mathbf{w}) = \mathbf{w} \cdot \mathbf{x}$ with fixed input $\mathbf{x}$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}$.

Differentiating with respect to w_1 gives:

$$\begin{aligned}\frac{\partial}{\partial w_1} (\bar{\mathbf{w}} \cdot \bar{\mathbf{x}}) &= \frac{\partial}{\partial w_1} \sum_{k=1}^n w_k \cdot x_k \\ &= \sum_{k=1}^n \frac{\partial}{\partial w_1} (w_k \cdot x_k) = \boxed{x_1} \\ &= \frac{\partial}{\partial w_1} (w_1 \cdot x_1) + \frac{\partial}{\partial w_1} (w_2 x_2) + \dots\end{aligned}$$

The pattern in the partial derivatives

We just saw that

$$\frac{\partial}{\partial w_1}(\mathbf{w} \cdot \mathbf{x}) = x_1,$$

The same reasoning applies to each parameter, so

$$\boxed{\frac{d}{d\mathbf{w}}(\mathbf{w} \cdot \mathbf{x}) = \mathbf{x}^T} \quad \frac{df}{d\mathbf{x}}$$

This is index notation: compute one general component, then identify the full derivative.

$$\begin{aligned} \frac{\partial}{\partial w_i} \left(\sum_{k=1}^n w_k \cdot x_k \right) & \quad \frac{d}{d\mathbf{w}} (\mathbf{w} \cdot \mathbf{x}) = [x_1, x_2, \dots, x_n] \\ & \quad = \mathbf{x}^T \\ & = \frac{\partial}{\partial w_i} (w_1 \cdot x_1) + \frac{\partial}{\partial w_i} (w_2 \cdot x_2) + \dots \\ & = \frac{\partial}{\partial w_i} (x_i \cdot w_i) = x_i \end{aligned}$$

Main tool: Kronecker delta

The Kronecker delta δ_{ij} checks whether two indices match:

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

$$\delta_{54} = 0$$

$$\delta_{11} = 1$$

How would you write $\frac{\partial x_i}{\partial x_j}$ using a Kronecker delta?

$$\frac{\partial x_i}{\partial x_j} = \delta_{ij}$$

What is $\sum_p \delta_{ip} u_p$?

$$= u_i$$

Going back - Kronecker delta filters

For $f(\mathbf{w}) = \mathbf{w} \cdot \mathbf{x}$ with fixed $\mathbf{x}$, find the derivative with respect to w_i :

$$\begin{aligned}\frac{\partial}{\partial w_i} \sum_{k=1}^n w_k x_k &= \sum_{k=1}^n \frac{\partial}{\partial w_i} (w_k x_k) \\ &= \sum_{k=1}^n x_k \cdot \frac{\partial w_k}{\partial w_i} \\ &= \sum_{k=1}^n x_k \cdot \delta_{ki} = x_i\end{aligned}$$

Linear algebra in index notation

$$\mathbf{v} \cdot \mathbf{v} = \sum_p v_p^2$$

$$[\mathbf{A}\mathbf{v}]_i = \sum_p A_{ip} v_p$$

$$[\mathbf{v}^T \mathbf{A}]_i = \sum_p v_p A_{pi}$$

$$[\mathbf{A}\mathbf{B}]_{ij} = \sum_p A_{ip} B_{pj}$$

$$\mathbf{v}^T \mathbf{A} \mathbf{u} = \sum_{p,q} v_p A_{pq} u_q$$

Example 1

Let $f: \mathbb{R}^m \rightarrow \mathbb{R}^n$ be given by

$$\frac{df}{d\bar{x}} \in \mathbb{R}^{n \times m}$$

$$\frac{d}{d\bar{x}} (A\bar{x}) = A$$

$$\left[\frac{df}{d\bar{x}} \right]_{ij} = \frac{\partial f_i}{\partial x_j}$$

$$f: x \mapsto Ax.$$

$$\downarrow \\ A \in \mathbb{R}^{n \times m}$$

$$x^T A$$

$$\frac{\partial}{\partial x_j} [A\bar{x}]_i = \frac{\partial}{\partial x_j} \sum_{k=1}^m A_{ik} x_k$$

$$= \sum_{k=1}^m \frac{\partial}{\partial x_j} A_{ik} x_k$$

$$= \sum_{k=1}^m A_{ik} \frac{\partial x_k}{\partial x_j} = \sum_{k=1}^m A_{ik} \delta_{kj} = \boxed{A_{ij}}$$

$$\boxed{\frac{df}{d\bar{x}} = A}$$

Example 2

Let $f: \mathbb{R}^m \rightarrow \mathbb{R}^n$ be given by

$$A \in \mathbb{R}^{m \times n}$$

$$J \in \mathbb{R}^{n \times m}$$

$$f: x \mapsto x^T A.$$

$$\frac{d}{dx} (x^T A) \stackrel{?}{=} A^T$$

$$\frac{\partial}{\partial x_j} [x^T A]_i = \frac{\partial}{\partial x_j} \sum_{k=1}^m x_k A_{ki}$$

$$= \sum_{k=1}^m A_{ki} \frac{\partial x_k}{\partial x_j}$$

$$= \sum_{k=1}^m A_{ki} \delta_{kj} = \boxed{A_{ji}} \Rightarrow$$

$$A_{ji} = [A^T]_{ij}$$

Example 3

$$|z|^2 = \sum_{i=1}^n z_i^2 = z^T z = z \cdot z \quad \frac{\partial}{\partial w_i} (\|X\bar{w} - \bar{y}\|^2)$$

$$f: w \mapsto \|Xw - y\|^2.$$

$$\begin{aligned} \|X\bar{w} - \bar{y}\|^2 &= (X\bar{w} - \bar{y})^T (X\bar{w} - \bar{y}) \\ &= (\bar{w}^T X^T - \bar{y}^T) (X\bar{w} - \bar{y}) \\ &= \bar{w}^T X^T X \bar{w} - \bar{y}^T X \bar{w} - \bar{w}^T X^T \bar{y} + \bar{y}^T \bar{y} \\ &= \bar{w}^T X^T X \bar{w} - 2 \underbrace{(\bar{y}^T X)}_{\text{red bracket}} \bar{w} \end{aligned}$$

$$\bar{y}^T X = z^T$$

$$2z^T \bar{w}$$

$$= 2z^T$$

$$= 2\bar{y}^T X$$

Example 3

$$\frac{\partial}{\partial \mathbf{w}} (\mathbf{w}^T \mathbf{X}^T \mathbf{y}) = \frac{d}{d\mathbf{w}} \left(\underbrace{\bar{\mathbf{w}} \cdot (\mathbf{X}^T \mathbf{y})}_{\mathbf{y}^T \mathbf{X}} \right)$$

$$\frac{\partial}{\partial w_i} \sum_p \sum_q w_p X_{qp} y_q \quad [\mathbf{y}^T \mathbf{X}]_i$$
$$\sum_p \sum_q X_{qp} y_q \delta_{pi} = \sum_q X_{qi} y_q \quad \uparrow$$

Example 3

$$S = X^T X$$

$$S^T = (X^T X)^T$$

$$= X^T X$$

$$\underline{2w^T X^T X}$$

$$\frac{\partial}{\partial w} (w^T X^T X w)$$

Write $S = X^T X$ to simplify this to $\frac{\partial}{\partial w} (w^T S w)$.

$$\begin{aligned} \frac{\partial}{\partial w_i} \sum_{p=1}^m \sum_{q=1}^m w_p S_{pq} w_q &= \sum_{p,q} S_{pq} \frac{\partial}{\partial w_i} (w_p w_q) \quad \nearrow \quad \frac{\partial w_p}{\partial w_i} w_q + \frac{\partial w_q}{\partial w_i} w_p \\ &= \sum_{p,q} S_{pq} (\delta_{pi} \cdot w_q + w_p \delta_{qi}) \quad \nearrow \quad [w^T S]_i \\ &= \sum_{p,q} S_{pq} \delta_{pi} w_q + \sum_{p,q} S_{pq} w_p \delta_{qi} = \sum_q S_{iq} w_q + \sum_p S_{pi} w_p \\ &= 2 w^T S \end{aligned}$$

Example 3

$$\frac{\partial}{\partial \mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 = 2\mathbf{w}^T \mathbf{X}^T \mathbf{X} - 2\mathbf{y}^T \mathbf{X}$$

Let's solve this!



$$2\mathbf{w}^T \mathbf{X}^T \mathbf{X} - 2\mathbf{y}^T \mathbf{X} = \mathbf{0}$$

$$\mathbf{w}^T \mathbf{X}^T \mathbf{X} = \mathbf{y}^T \mathbf{X}$$

$$\mathbf{X}^T \mathbf{X} \bar{\mathbf{w}} = \mathbf{X}^T \bar{\mathbf{y}}$$

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Questions?

Ask me anything!

$$\sum_q S_{iq} w_q + \underbrace{\sum_p S_{pi} w_p}_{\downarrow}$$
$$\downarrow$$
$$[S w]_i \quad [w^T S]_i$$

$$S = X^T X$$